

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, SQL, C++, Bash

Tools/Frameworks: Docker, FastAPI, Flask, TensorFlow, PyTorch, Spark, Databricks, Kubernetes, Airflow, Kafka, dbt

Cloud Platforms: AWS, GCP, Azure

Competence: Machine Learning, MLOps, Data Engineering, ETL Pipelines, Model Deployment, Data Visualization, AI Architecture Design

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated deep learning models for metrological applications, enhancing model performance and training efficiency.
- Developed and tested physics-informed hybrid loss functions to improve model accuracy and stability.
- Collaborated with PhD researchers to co-author a publication presented at the FLOMEKO conference.
- Built interactive dashboards for data visualization and real-time model performance monitoring.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Increased vitiligo segmentation accuracy to 94% and improved processing speed by 5x using advanced AI algorithms, deployed via Docker and FastAPI.
- Developed a real-time tennis foul detection system with TensorRT, reducing compute costs and time by 6x.
- Managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift, and maintained Kubernetes infrastructure.
- Led educational sessions on machine learning and big data technologies at universities, mentoring over 50 students.
- Created proof-of-concepts using YOLO, few-shot learning, and CNNs for client demonstrations.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed a full-stack web application providing AI-based services using Flask, HTML, and CSS.

- Built an AI chatbot application leveraging OpenAI API and large language model (LLM) frameworks.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Developed automated ETL pipelines for structured and unstructured data, optimizing data ingestion workflows.
- Built web scrapers to extract data from diverse sources, enabling actionable client analytics.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Designed a weighted-probability ensemble integrating DPFunc and DeepGOPlus, leveraging AlphaFold for feature extraction.
- Improved model accuracy by 4% through custom feature engineering across MF, CC, and BP namespaces.

Razanalytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed a real-time Bitcoin price forecasting platform using LSTM networks and time-series analysis.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Built the world's first AI-driven self-playing Chitralli Sitar, combining signal processing and music generation LLMs.
- Project featured on Coke Studio in a music video by Hassan Raheem, showcasing innovative AI applications in music.

Hugging Face

Open-Source

Contributor

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specializing in Machine Learning, MLOps, and Data Engineering.

National University of Computer and Emerging Sciences

Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor for academic excellence.
- Focused on Artificial Intelligence, Data Science, and Software Development.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python